

Lines and Queries II

Time limit: 1.00 s

Memory limit: 512 MB

Your task is to efficiently process the following types of queries:

1. Add a line $ax + b$ that is active in range $[l, r]$
2. Find the maximum point in any active line at position x

Input

The first line has an integer n : the number of queries.

The following n lines describe the queries. The format of each line is either "1 a b l r " or "2 x ".

Output

Print the answer for each query of type 2. If no line is active, print NO.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$
- $-10^9 \leq a, b \leq 10^9$
- $0 \leq x \leq 10^5$
- $0 \leq l \leq r \leq 10^5$

Example

Input	Output
6 1 1 2 1 3 2 3 2 4 1 0 4 1 5 2 3 2 4	5 NO 5 4